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Inventor(s)	Hur; Joonhoi et al.

Electronic devices with angular location detection capabilities

Abstract

An electronic device may include wireless circuitry having a set of two or more antennas coupled to voltage standing wave ratio (VSWR) sensors. The VSWR sensors may gather VSWR measurements from radio-frequency signals transmitted using the set of antennas. The antennas may be disposed on one or more substrates and/or may be formed from conductive portions of a housing. Control circuitry may process the VSWR measurements to identify the ranges between each of the antennas in the set of antennas and an external object. The control circuitry may process the ranges to identify an angular location of the external object with respect to the device. The control circuitry may adjust subsequent communications based, adjust the direction of a signal beam produced by a phased antenna array, identify a user input, or perform any other desired operations based on the angular location.

Inventors: **Hur; Joonhoi (Sunnyvale, CA), Menkhoff; Andreas (Oberhaching, DE), Sogl; Bernhard (Unterhaching, DE), Schrattenecker; Jochen (Alberndorf in der Riedmark, AT), Vazny; Rastislav (Saratoga, CA)**

Applicant: **Apple Inc.** (Cupertino, CA)

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Primary Examiner: Moore; Whitney

Attorney, Agent or Firm: Treyz Law Group, P.C.

Background/Summary

(1) This application is a continuation of U.S. patent application Ser. No. 17/331,504, filed May 26, 2021, which is hereby incorporated by reference herein in its entirety.

FIELD
(1) This disclosure relates generally to electronic devices and, more particularly, to electronic devices with wireless circuitry.
BACKGROUND
(2) Electronic devices are often provided with wireless capabilities. An electronic device with wireless capabilities has wireless circuitry that includes one or more antennas. The wireless circuitry is sometimes used to perform spatial ranging operations in which radio-frequency signals are used to estimate a distance between the electronic device and external objects.
(3) It can be challenging to provide wireless circuitry that accurately estimates this distance. For

example, the wireless circuitry will often exhibit a blind spot near the device within which the wireless circuitry is unable to accurately detect the presence of external objects. In addition, it can be difficult for the wireless circuitry to fully characterize the location and orientation of external objects when present within the blind spot.

SUMMARY

(4) An electronic device may include wireless circuitry controlled by one or more processors. The wireless circuitry may include a set of two or more antennas communicably coupled to voltage standing wave ratio (VSWR) sensors. The VSWR sensors may gather VSWR measurements from radio-frequency signals transmitted using the set of antennas. The antennas in the set of antennas may be disposed on one or more substrates and/or may be formed from conductive portions of a housing for the device. One or more processors may process the VSWR measurements to identify the ranges between each of the antennas in the set of antennas and an external object at, adjacent, or proximate to the set of antennas. The one or more processors may process the ranges to identify an angular location of the external object with respect to the device.

(5) The one or more processors may perform any desired operations based on the identified angular location. For example, the one or more processors may adjust subsequent communications by one or more of the antennas based on the angular location (e.g., by reducing a maximum transmit power level of one or more of the antennas). If desired, the one or more processors may adjust the direction of a signal beam produced by a phased antenna array based on the angular location (e.g., to steer the signal beam around the external object). As another example, the one or more processors may identify a user input or gesture based on the angular location.

(6) An aspect of the disclosure provides an electronic device operable in an environment that includes an external object. The electronic device can include a first antenna and a second antenna. The electronic device can include a first voltage standing wave ratio (VSWR) sensor communicably coupled to the first antenna. The first VSWR sensor can be configured to perform a first VSWR measurement using radio-frequency signals transmitted by the first antenna. The electronic device can include a second VSWR sensor communicably coupled to the second antenna. The second VSWR sensor can be configured to perform a second VSWR measurement using radio-frequency signals transmitted by the second antenna. The electronic device can include one or more processors. The one or more processors can be configured to identify a first range from the first antenna to the external object based on the first VSWR measurement. The one or more processors can be configured to identify a second range from the second antenna to the external object based on the second VSWR measurement. The one or more processors can be configured to identify an angular location of the external object based at least on the first range and the second range.

(7) An aspect of the disclosure provides a method for operating an electronic device having a set of antennas, at least one voltage standing wave ratio (VSWR) sensor communicably coupled to the set of antennas, and one or more processors. The set of antennas can include at least two antennas. The method can include with the set of antennas, transmitting radio-frequency signals. The method can include with the at least one VSWR sensor, gathering VSWR measurements from the radio-frequency signals transmitted by different antennas in the set of antennas. The method can include with the one or more processors, identifying a plurality of ranges between the set of antennas and the external object based on the VSWR measurements. The method can include with the one or more processors, identifying an angular location of the external object based on the plurality of ranges between the set of antennas and the external object.

(8) An aspect of the disclosure provides a method of operating an electronic device in an environment having an external object. The method can include with a first antenna on the electronic device, transmitting first radio-frequency signals. The method can include with a second antenna on the electronic device, transmitting second radio-frequency signals. The method can include with a first voltage standing wave ratio (VSWR) sensor communicably coupled to the first

antenna, gathering a first VSWR measurement using the first radio-frequency signals transmitted using the first antenna. The method can include with a second VSWR sensor communicably coupled to the second antenna, gathering a second VSWR measurement using the second radio-frequency signals transmitted using the second antenna. The method can include with one or more processors, identifying an angular location of the external object based at least on the first VSWR measurement and the second VSWR measurement. The method can include with the one or more processors, adjusting a subsequent transmission by the first antenna based at least on the angular location of the external object.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a functional block diagram of an illustrative electronic device having voltage standing wave ratio (VSWR) sensors for detecting the angular location of an external object in accordance with some embodiments.

(2) FIG. **2** is a circuit diagram of an illustrative VSWR sensor having a directional coupler for detecting the range between an external object and an antenna in accordance with some embodiments.

(3) FIG. **3** is a plot of reflection coefficient as a function of frequency that may be produced by an illustrative VSWR sensor for detecting the range between an external object and an antenna in accordance with some embodiments.

(4) FIG. **4** is a plot showing how the reflection coefficient measured by an illustrative VSWR sensor may vary at different times when external objects are present at different ranges from an antenna in accordance with some embodiments.

(5) FIG. **5** is a plot showing how reflection coefficient variation may be correlated to the range between an antenna and an external object in accordance with some embodiments.

(6) FIG. **6** is a perspective view showing how an external object may be present at a given angular location over an electronic device surface in accordance with some embodiments.

(7) FIG. **7** is a flow chart of illustrative operations involved in detecting the angular location of an external object using VSWR sensors and multiple antennas in accordance with some embodiments.

(8) FIG. **8** is a top view showing how multiple antennas may be used to detect the angular location of an external object in accordance with some embodiments.

(9) FIG. **9** is a side view showing how multiple antennas may be used to detect the angular location of an external object in accordance with some embodiments.

(10) FIG. **10** is a side view showing how multiple antennas may perform beam steering operations based on the detected angular location of an external object in accordance with some embodiments.

(11) FIG. **11** is a top view showing how antennas used to detect the angular location of an external object may be distributed across multiple arrays at different orientations in accordance with some embodiments.

(12) FIG. **12** is a top view showing how antennas used to detect the angular location of an external object may be distributed across an electronic device in accordance with some embodiments.

DETAILED DESCRIPTION

(13) Electronic device **10** of FIG. **1** may be a computing device such as a laptop computer, a desktop computer, a computer monitor containing an embedded computer, a tablet computer, a cellular telephone, a media player, or other handheld or portable electronic device, a smaller device such as a wristwatch device, a pendant device, a headphone or earpiece device, a device embedded in eyeglasses or other equipment worn on a user's head, or other wearable or miniature device, a television, a computer display that does not contain an embedded computer, a gaming device, a navigation device, an embedded system such as a system in which electronic equipment with a

display is mounted in a kiosk or automobile, a wireless internet-connected voice-controlled speaker, a home entertainment device, a remote control device, a gaming controller, a peripheral user input device, a wireless base station or access point, equipment that implements the functionality of two or more of these devices, or other electronic equipment.

(14) As shown in the functional block diagram of FIG. **1**, device **10** may include components located on or within an electronic device housing such as housing **12**. Housing **12**, which may sometimes be referred to as a case, may be formed of plastic, glass, ceramics, fiber composites, metal (e.g., stainless steel, aluminum, metal alloys, etc.), other suitable materials, or a combination of these materials. In some situations, parts or all of housing **12** may be formed from dielectric or other low-conductivity material (e.g., glass, ceramic, plastic, sapphire, etc.). In other situations, housing **12** or at least some of the structures that make up housing **12** may be formed from metal elements.

(15) Device **10** may include control circuitry **14**. Control circuitry **14** may include storage such as storage circuitry **16**. Storage circuitry **16** may include hard disk drive storage, nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Storage circuitry **16** may include storage that is integrated within device **10** and/or removable storage media.

(16) Control circuitry **14** may include processing circuitry such as processing circuitry **18**. Processing circuitry **18** may be used to control the operation of device **10**. Processing circuitry **18** may include on one or more microprocessors, microcontrollers, digital signal processors, host processors, baseband processor integrated circuits, application specific integrated circuits, central processing units (CPUs), graphics processing units (GPUs), etc. Control circuitry **14** may be configured to perform operations in device **10** using hardware (e.g., dedicated hardware or circuitry), firmware, and/or software. Software code for performing operations in device **10** may be stored on storage circuitry **16** (e.g., storage circuitry **16** may include non-transitory (tangible) computer readable storage media that stores the software code). The software code may sometimes be referred to as program instructions, software, data, instructions, or code. Software code stored on storage circuitry **16** may be executed by processing circuitry **18**. If desired, portions of storage circuitry **16** may be located on processing circuitry **18** (e.g., as L1 and L2 cache), whereas other portions of storage circuitry **16** are located external to processing circuitry **18** (e.g., while remaining accessible to processing circuitry **18** via a memory interface).

(17) Control circuitry **14** may be used to run software on device **10** such as satellite navigation applications, internet browsing applications, voice-over-internet-protocol (VOIP) telephone call applications, email applications, media playback applications, operating system functions, etc. To support interactions with external equipment, control circuitry **14** may be used in implementing communications protocols. Communications protocols that may be implemented using control circuitry **14** include internet protocols, wireless local area network (WLAN) protocols (e.g., IEEE 802.11 protocols—sometimes referred to as Wi-Fi®), protocols for other short-range wireless communications links such as the Bluetooth® protocol or other wireless personal area network (WPAN) protocols, IEEE 802.11ad protocols (e.g., ultra-wideband protocols), cellular telephone protocols (e.g., 3G protocols, 4G (LTE) protocols, 3GPP Fifth Generation (5G) protocols, etc.), antenna diversity protocols, satellite navigation system protocols (e.g., global positioning system (GPS) protocols, global navigation satellite system (GLONASS) protocols, etc.), antenna-based spatial ranging protocols (e.g., radio detection and ranging (RADAR) protocols or other desired range detection protocols for signals conveyed at millimeter and centimeter wave frequencies), or any other desired communications protocols. Each communications protocol may be associated with a corresponding radio access technology (RAT) that specifies the physical connection methodology used in implementing the protocol.

(18) Device **10** may include input-output circuitry **20**. Input-output circuitry **20** may include input-

output devices **22**. Input-output devices **22** may be used to allow data to be supplied to device **10** and to allow data to be provided from device **10** to external devices. Input-output devices **22** may include user interface devices, data port devices, and other input-output components. For example, input-output devices **22** may include touch sensors, displays (e.g., touch-sensitive and/or force-sensitive displays), light-emitting components such as displays without touch sensor capabilities, buttons (mechanical, capacitive, optical, etc.), scrolling wheels, touch pads, key pads, keyboards, microphones, cameras, buttons, speakers, status indicators, audio jacks and other audio port components, digital data port devices, motion sensors (accelerometers, gyroscopes, and/or compasses that detect motion), capacitance sensors, temperature sensors, proximity sensors, magnetic sensors, force sensors (e.g., force sensors coupled to a display to detect pressure applied to the display), etc. In some configurations, keyboards, headphones, displays, pointing devices such as trackpads, mice, and joysticks, and other input-output devices may be coupled to device **10** using wired or wireless connections (e.g., some of input-output devices **22** may be peripherals that are coupled to a main processing unit or other portion of device **10** via a wired or wireless link).

(19) Input-output circuitry **20** may include wireless circuitry **24** to support wireless communications and/or radio-based spatial ranging operations. Wireless circuitry **24** may include two or more antennas **40**. Wireless circuitry **24** may also include baseband processor circuitry, transceiver circuitry, amplifier circuitry, filter circuitry, switching circuitry, analog-to-digital converter (ADC) circuitry, digital-to-analog converter (DAC) circuitry, radio-frequency transmission lines, and/or any other circuitry for transmitting and/or receiving radio-frequency signals using antennas **40**.

(20) Antennas **40** may be formed using any desired antenna structures. For example, antennas **40** may include antennas with resonating elements that are formed from loop antenna structures, patch antenna structures, inverted-F antenna structures, slot antenna structures, planar inverted-F antenna structures, helical antenna structures, monopole antennas, dipoles, hybrids of these designs, etc. Filter circuitry, switching circuitry, impedance matching circuitry, and/or other antenna tuning components may be adjusted to adjust the frequency response and wireless performance of antennas **40** over time.

(21) Wireless circuitry **24** may use antennas **40** to transmit and/or receive radio-frequency signals **38** to convey wireless communications data between device **10** and external wireless communications equipment **28** (e.g., one or more other devices such as device **10**, a wireless access point or base station, etc.). Wireless communications data may be conveyed by wireless circuitry **24** bidirectionally or unidirectionally. The wireless communications data may, for example, include data that has been encoded into corresponding data packets such as wireless data associated with a telephone call, streaming media content, internet browsing, wireless data associated with software applications running on device **10**, email messages, etc.

(22) Wireless circuitry **24** may include communications and/or long range spatial ranging circuitry **26** (sometimes referred to herein simply as communications circuitry **26**). Communications circuitry **26** may transmit and/or receive wireless communications data using antennas **40**. Communications circuitry **26** may include baseband circuitry (e.g., one or more baseband processors) and one or more radios (e.g., radios having radio-frequency transceivers, modems, synthesizers, switches, filters, mixers, ADCs, DACs, amplifiers, etc.) for conveying radio-frequency signals **38** using one or more antennas **40**.

(23) Communications circuitry **26** may transmit and/or receive radio-frequency signals **38** within a corresponding frequency band at radio frequencies (sometimes referred to herein as a communications band or simply as a “band”). The frequency bands handled by communications circuitry **26** may include wireless local area network (WLAN) frequency bands (e.g., Wi-Fi® (IEEE 802.11) or other WLAN communications bands) such as a 2.4 GHz WLAN band (e.g., from 2400 to 2480 MHz), a 5 GHz WLAN band (e.g., from 5180 to 5825 MHz), a Wi-Fi® 6E band (e.g., from 5925-7125 MHz), and/or other Wi-Fi® bands (e.g., from 1875-5160 MHz), wireless personal

area network (WPAN) frequency bands such as the 2.4 GHz Bluetooth® band or other WPAN communications bands, cellular telephone frequency bands (e.g., bands from about 600 MHz to about 5 GHz, 3G bands, 4G LTE bands, 5G New Radio Frequency Range 1 (FR1) bands below 10 GHz, 5G New Radio Frequency Range 2 (FR2) bands between 20 and 60 GHz, etc.), other centimeter or millimeter wave frequency bands between 10-300 GHz, near-field communications frequency bands (e.g., at 13.56 MHz), satellite navigation frequency bands (e.g., a GPS band from 1565 to 1610 MHz, a Global Navigation Satellite System (GLONASS) band, a BeiDou Navigation Satellite System (BDS) band, etc.), ultra-wideband (UWB) frequency bands that operate under the IEEE 802.15.4 protocol and/or other ultra-wideband communications protocols, communications bands under the family of 3GPP wireless communications standards, communications bands under the IEEE 802.XX family of standards, and/or any other desired frequency bands of interest.

(24) Communications circuitry **26** may be coupled to antennas **40** using one or more transmit paths **34** and/or one or more receive paths **36**. Communications circuitry **26** may uses transmit paths **34** to transmit radio-frequency signals **38** and may use receive paths **36** to receive radio-frequency signals **38**. Transmit paths **34** (sometimes referred to herein as transmit chains **34**) may include one or more signal paths (e.g., radio-frequency transmission lines), amplifier circuitry, filter circuitry, switching circuitry, radio-frequency front end circuitry (e.g., components on a radio-frequency front end module), and/or any other desired paths or circuitry for transmitting radio-frequency signals from communications circuitry **26** to antenna(s) **40**. Receive paths **36** may include one or more signal paths (e.g., radio-frequency transmission lines), amplifier circuitry (e.g., low noise amplifier (LNA) circuitry), filter circuitry, switching circuitry, radio-frequency front end circuitry (e.g., components on a radio-frequency front end module), and/or any other desired paths or circuitry for conveying radio-frequency signals from antenna(s) **40** to communications circuitry **26**.

(25) In addition to conveying wireless communications data, communications circuitry **26** may additionally or alternatively use antennas **40** to perform long range spatial ranging operations. Communications circuitry **26** may include long range spatial ranging circuitry for performing long range spatial ranging operations. The long range spatial ranging circuitry in communications circuitry **26** may include mixer circuitry, amplifier circuitry, transmitter circuitry (e.g., signal generators, synthesizers, etc.), receiver circuitry, filter circuitry, baseband circuitry, ADC circuitry, DAC circuitry, and/or any other desired components used in performing spatial ranging operations using antennas **40**. The long range spatial ranging circuitry may include, for example, radar circuitry (e.g., frequency modulated continuous wave (FMCW) radar circuitry, OFDM radar circuitry, FSCW radar circuitry, a phase coded radar circuitry, other types of radar circuitry). Antennas **40** may include separate antennas for conveying wireless communications data and radio-frequency signals for spatial ranging or may include one or more antennas **40** that are used to both convey wireless communications data and to perform spatial ranging. Using a single antenna **40** to both convey wireless communications data and perform spatial ranging may, for example, serve to minimize the amount of space occupied in device **10** by antennas **40**.

(26) When performing long range spatial ranging operations, the long range spatial ranging circuitry in communications circuitry **26** may use a first antenna **40** (e.g., a transmit antenna) to transmit radio-frequency signals **42**. Radio-frequency signals **42** may include one or more signal tones, continuous waves of radio-frequency energy, wideband signals, chirp signals, or any other desired transmit signals (e.g., radar signals) for use in spatial ranging operations. Unlike radio-frequency signals **38**, radio-frequency signals **42** may be free from wireless communications data (e.g., cellular communications data packets, WLAN communications data packets, etc.). Radio-frequency signals **42** may sometimes also be referred to herein as spatial ranging signals **42**, long range spatial ranging signals **42**, or radar signals **42**. The long range spatial ranging circuitry in communications circuitry **26** may transmit radio-frequency signals **42** at one or more carrier frequencies in a corresponding radio frequency band such (e.g., a frequency band that includes frequencies greater than around 10 GHz, greater than around 20 GHz, less than 10 GHz, 20-30

GHz, greater than 40 GHz, etc.).

(27) Radio-frequency signals **42** may reflect off of objects external to device **10** such as external object **46**. External object **46** may be, for example, the ground, a building, part of a building, a wall, furniture, a ceiling, a person, a body part, an animal, a vehicle, a landscape or geographic feature, an obstacle, external communications equipment such as external wireless communications equipment **28**, another device of the same type as device **10** or a peripheral device such as a gaming controller or remote control, or any other physical object or entity that is external to device **10**. A second antenna **40** (e.g., a receive antenna) in wireless circuitry **24** may receive reflected radio-frequency signals **44**. Reflected signals **44** may be a reflected version of the transmitted radio-frequency signals **42** that have reflected off of external object **46** and back towards device **10**.

(28) The long range spatial ranging circuitry in communications circuitry **26** may receive reflected signals **44** from the second antenna **40** via a corresponding receive path **36**. Control circuitry **14** may process the transmitted radio-frequency signals **42** and the received reflected signals **44** to detect or estimate the range R between device **10** and external object **46**. If desired, control circuitry **14** may also process the transmitted and received signals to identify a two or three-dimensional spatial location (position) of external object **46**, a velocity of external object **46**, and/or an angle of arrival of reflected signals **44**. If desired, a loopback path may be coupled between the transmit path **34** and the receive path **36** used by the long range spatial ranging circuitry. The loopback path may be used to convey transmit signals on the transmit path to receiver circuitry in the long range spatial ranging circuitry. As an example, in embodiments where the long range spatial ranging circuitry performs spatial ranging using an FMCW scheme, the loopback path may be a de-chirp path that conveys chirp signals on the transmit path to a de-chirp mixer in the long range spatial ranging circuitry. In these embodiments, doppler shifts in continuous wave transmit signals may be detected and processed to identify the velocity of external object **46**, and the time dependent frequency difference between radio-frequency signals **42** and reflected signals **44** may be detected and processed to identify range R and/or the position of external object **46**. Use of continuous wave signals for estimating range R may allow control circuitry **14** to reliably distinguish between external object **46** and other background or slower-moving objects, for example. This example is merely illustrative and, in general, the long range spatial ranging circuitry may implement any desired radar or long range spatial ranging scheme.

(29) The radio-frequency transmission lines in transmit paths **34** and receive paths **36** may include coaxial cables, microstrip transmission lines, stripline transmission lines, edge-coupled microstrip transmission lines, edge-coupled stripline transmission lines, transmission lines formed from combinations of transmission lines of these types, etc. Transmission lines in device may be integrated into rigid and/or flexible printed circuit boards if desired. One or more radio-frequency lines may be shared between transmit path(s) **34** and receive path(s) **36** if desired. The components of wireless circuitry **24** may be formed on one or more common substrates or modules (e.g., rigid printed circuit boards, flexible printed circuit boards, integrated circuits, chips, packages, systems-on-chip, etc.).

(30) The example of FIG. **1** is merely illustrative. While control circuitry **14** is shown separately from wireless circuitry **24** in the example of FIG. **1** for the sake of clarity, wireless circuitry **24** may include processing circuitry that forms a part of processing circuitry **18** and/or storage circuitry that forms a part of storage circuitry **16** of control circuitry **14** (e.g., portions of control circuitry **14** may be implemented on wireless circuitry **24**). As an example, some or all of the baseband circuitry in communications circuitry **26** may form a part of control circuitry **14**. The baseband processor circuitry may, for example, access a communication protocol stack on control circuitry **14** (e.g., storage circuitry **20**) to: perform user plane functions at a PHY layer, MAC layer, RLC layer, PDCP layer, SDAP layer, and/or PDU layer, and/or to perform control plane functions at the PHY layer, MAC layer, RLC layer, PDCP layer, RRC, layer, and/or non-access stratum layer. If desired, the PHY layer operations may additionally or alternatively be performed by radio-frequency (RF)

interface circuitry in wireless circuitry **24**. In addition, wireless circuitry **24** may include any desired number of antennas **40**. Each antenna **40** may be coupled to communications circuitry **26** over dedicated transmit and/or receive paths or over one or more transmit and/or receive paths that are shared between antennas. Communications circuitry **26** may convey wireless communications data without performing spatial ranging operations (e.g., the long range spatial ranging circuitry in communications circuitry **26** may be omitted) or communications circuitry **26** may perform spatial ranging operations without conveying wireless communications data.
(31) The long range spatial ranging circuitry in communications circuitry **26** may be used to accurately identify range R when external object **46** is at relatively far distances from device **10**. However, in practice, the long range spatial ranging circuitry exhibits a blind spot to nearby external objects at distances less than threshold range R.sub.TH (e.g., around 1-2 cm) from device **10**. When external object **46** is located within this blind spot (e.g., within threshold range R.sub.TH from transmit antenna **40**TX), the long range spatial ranging circuitry may be unable to identify the presence, location, and/or velocity of external object **46** with a satisfactory level of accuracy. External objects **46** within threshold range R.sub.TH of antenna(s) **40** may be exposed to relatively high amounts of radio-frequency energy (e.g., from the radio-frequency signals **38** and/or **42** that are transmitted by antenna(s) **40**). In scenarios where external object **46** is a body part or person, if care is not taken, this transmitted radio-frequency energy may cause wireless circuitry **24** to exceed regulatory limits or other limits on specific absorption rate (SAR) (e.g., when the transmitted signals are at frequencies below 6 GHz) and/or maximum permissible exposure (MPE) (e.g., when the transmitted signals are at frequencies above 6 GHz). In order to detect the presence of external object **46** within threshold range R.sub.TH from antenna(s) **40**, wireless circuitry **24** may include an ultra-short range (USR) object detector such as USR detector **30**. USR detector **30** may serve to detect external object **46** at ultra-short ranges (e.g., at ranges within threshold range R.sub.TH from antenna(s) **40**). In other words, USR detector **30** may perform external object detection within the blind spot of the long range spatial ranging circuitry in communications circuitry **26**.
(32) USR detector **30** may include two or more voltage standing wave ratio (VSWR) sensors (detectors) such as VSWR sensors **32**. Each VSWR sensor **32** may be interposed on a respective transmit path **34**. Each VSWR sensor **32** may gather VSWR values using the antenna **40** coupled to its respective transmit path **34**. The VSWR values may include complex scattering parameter values (S-parameter values) such as reflection coefficient (return loss) values (e.g., S.sub.11 values). The magnitude of the S.sub.11 values (e.g., |S.sub.11| values) may be indicative of the amount of transmitted radio-frequency energy that is reflected in a reverse direction along the transmit path (e.g., in response to the presence of external object **46** at or adjacent to the corresponding antenna **40**). The VSWR values gathered by each VSWR sensor **32** may be insensitive to situations where external object **46** is located at distances greater than threshold range R.sub.TH from antenna(s) **40**. However, the VSWR values gathered by VSWR sensors **32** may allow control circuitry **14** to identify when external object **46** is located within threshold range R.sub.TH from two or more of the antennas **40** in wireless circuitry **24** (e.g., within the blind spot of the long range spatial ranging circuitry in communications circuitry **26**).
(33) In this way, USR detector **30** and the long range spatial ranging circuitry may identify the presence of external object **46** and optionally the range R to external object **46**, regardless of whether external object **46** has moved to a position that is relatively close or relatively far from device **10** over time. In addition, USR detector **30** may identify the presence of external object **46** within the blind spot of the long range spatial ranging circuitry in communications circuitry **26** so that suitable action can be taken to ensure that wireless circuitry **24** continues to satisfy any applicable SAR and/or MPE regulations. By using the same antenna(s) **40** to both transmit radio-frequency signals **38**/**42** and measure VSWR, the VSWR measurements will be very closely correlated with the amount of radio-frequency energy absorbed by external object **46** from the transmitted radio-frequency signals **38**/**42**, thereby providing high confidence in the use of USR

detector **30** for meeting any applicable SAR and/or MPE regulations (e.g., greater confidence than in scenarios where proximity sensors that are separate from the transmit antenna or transmit chain are used to identify the presence of external objects within threshold range R.sub.TH of device **10**).
(34) In the example of FIG. **1**, two antennas **40** are illustrated as being communicable coupled to respective VSWR sensors **32** in USR detector **30**. In general, a set of any desired number N of two or more antennas **40** may be communicably coupled to a respective VSWR sensor **32** (e.g., VSWR sensors **32** may be disposed/interposed on any desired number of two or more of the transmit paths **34** in wireless circuitry **24**). All of the antennas **40** may have a corresponding VSWR sensor **32** or only a subset of the antennas **40** may have a corresponding VSWR sensor **32**. By using more than one antenna **40** to gather (perform) VSWR measurements, control circuitry **14** may process the VSWR measurements to identify the range R between external object **46** and each antenna **40** having a VSWR sensor **32**. Control circuitry **14** may process the range R between external object **46** and each of the antennas **40** having a VSWR sensor **32** to identify the location (e.g., the angular location) of external object **46** relative to a surface of device **10**. Control circuitry **14** may use the identified angular location of external object **46** to perform any desired processing tasks, such as to perform beam steering using a phased antenna array of antennas **40** (e.g., to steer around external object **46**), to identify a user input or gesture corresponding to the angular location of external object **46**, to adjust the maximum transmit power level for one or more antennas **40**, etc.
(35) FIG. **2** is a circuit diagram of one of the VSWR sensors **32** in wireless circuitry **24** disposed on a corresponding transmit path **34**. As shown in FIG. **2**, transmit path **34** may include a power amplifier (PA) such as PA **96**. The input of PA **96** may be coupled to communications circuitry **26** of FIG. **1**. The output of PA **96** may be coupled to a corresponding antenna **40** via a switch such as antenna switch **94**. The output of PA **96** may also be coupled to matched load **88** via a switch such as matched load switch **90**. Matched load **88** may be coupled in series between matched load switch **90** and ground **82**. Matched load **88**, matched load switch **90**, and/or antenna switch **94** may be omitted if desired.
(36) In the example of FIG. **2**, VSWR sensor **32** is a directional switch coupler. This is merely illustrative and, in general, VSWR sensor **32** may be implemented using any desired VSWR sensor architecture. As shown in FIG. **2**, VSWR sensor **32** may include directional coupler **72** interposed on transmit path **34** between PA **96** and antenna **40** (e.g., along a radio-frequency transmission line in transmit path **34** coupled between the output of PA **96** and antenna **40**). Directional coupler **72** may have a first port (P**1**) coupled to the output of PA **96** and a second port (P**2**) communicably coupled to antenna **40**. Directional coupler **72** may have a third port (P**3**) coupled to a first termination that includes resistor **84** coupled in series between termination switch **78** and ground **82**. Directional coupler **72** may also have a fourth port (P**4**) coupled to a second termination that includes resistor **86** coupled in series between termination switch **80** and ground **82**. VSWR sensor **32** may have a forward (FW) switch **74** coupled between port P**3** and measurement circuitry **70** (e.g., an amplitude and/or phase detector). VSWR sensor **32** may also have a reverse (RW) switch **76** coupled between port P**4** and measurement circuitry **70**.
(37) Measurement circuitry **70** may have a control path coupled to other components in USR detector **30** or control circuitry **14** (FIG. **1**) and/or some or all of measurement circuitry **70** may form a part of control circuitry **14** (e.g., the operations of some or all of measurement circuitry **70** may be performed using one or more processors). Measurement circuitry **70** may include, for example, a power detector such as power detector **98**, an in-phase and quadrature-phase (I/Q) detector (e.g., an ADC), logic such as comparator/logic **102** (e.g., one or more logic gates, etc.), and/or memory such as memory **104**. Memory **104** may form a part of storage circuitry **16** of FIG. **1**, for example. If desired, I/Q detector **100** may be formed from one or more ADCs in one of the receive paths **36** of wireless circuitry **24** (FIG. **1**).
(38) When gathering (performing) VSWR measurements (e.g., S-parameter values such as S.sub.11 values), PA **96** may output a transmit test signal sigtx (e.g., while antenna switch **94** is closed). Test

signal sigtx may be a radar transmit signal transmitted by long range spatial ranging circuitry in communications circuitry **26** (e.g., radio-frequency signals **42** of FIG. **1**), a wireless communications data transmit signal transmitted by communications circuitry **26** (e.g., radio-frequency signals **38** of FIG. **1**), or a dedicated test signal for use in VSWR measurement (e.g., one or more tones transmitted by a signal generator, local oscillator, and/or other signal generation circuitry in USR detector **30** of FIG. **1**). For example, a sequential signal generator **108** may be used to generate test signal sigtx. Sequential signal generator **108** may be a part of the long range spatial ranging circuitry in communications circuitry **26** (e.g., test signal sigtx may be a continuous wave or wideband that can also be used in performing long range spatial ranging operations), may be a part of a transceiver in communications circuitry **26** that transmits wireless communications data (e.g., test signal sigtx may also carry wireless communications data), or may be formed as a part of USR detector **30** that is separate from communications circuitry **26**. Additionally or alternatively, a simple local oscillator such as local oscillator (LO) **106** may generate test signal sigtx.

(39) In performing VSWR measurements, VSWR sensor **32** may perform forward path measurements and reverse path measurements using transmit signal sigtx. When performing forward path measurements, FW switch **74** is closed, RW switch **76** is open, switch **80** is closed, and switch **78** is open so that test signal sigtx is coupled off from transmit path **34** by directional coupler **72** and routed to measurement circuitry **70** through FW switch **74**. Measurement circuitry **70** may measure and store the amplitude (magnitude) and/or phase of test signal sigtx for further processing (e.g., as a forward signal phase and magnitude measurement). For example, power detector **98** (e.g., a peak detector, diode and capacitor, etc.) may measure the magnitude of test signal sigtx and may store the magnitude on memory **104**. As another example, I/Q detector **100** may make I/Q measurements for the forward path that are stored on memory **104**.

(40) At least some of test signal sigtx will reflect off of antenna **40** (e.g., due to impedance discontinuities between transmit path **34** and antenna **40**, subject to impedance loading from any external objects at or adjacent to antenna **40**) and back towards PA **96** as reflected test signal sigtx′. When performing reverse path measurements, FW switch **74** is open, RW switch **76** is closed, switch **80** is open, and switch **78** is closed so that reflected test signal sigtx′ is coupled off of transmit path **34** by directional coupler **72** and routed to measurement circuitry **70** through RW switch **76**. Measurement circuitry **70** (e.g., power detector **98** or I/Q detector **100**) may measure and store the amplitude (magnitude) and/or phase of reflected test signal sigtx′ for further processing (e.g., as a reverse signal phase and magnitude measurement). Comparator/logic **102** and/or control circuitry **14** (FIG. **1**) may process the stored forward and reverse phase and magnitude measurements to identify complex scattering parameter values such as S.sub.11 values. The S.sub.11 values are characterized by a scalar magnitude |S.sub.11| and a corresponding phase. In this way, VSWR sensor **32** may measure VSWR values (e.g., S.sub.11 values, |S.sub.11| values, etc.) that can be used to determine when external object **46** is located at a range R that is less than or equal to threshold range R.sub.TH. Long range spatial ranging circuitry in communications circuitry **26** (FIG. **1**) may also use antenna **40** to identify range R when external object **46** is located at a range R that is beyond threshold range R.sub.TH from antenna **40**.

(41) If desired, control circuitry **14** may compare the VSWR measurements to one or more threshold values to identify range R. FIG. **3** is a plot showing how VSWR measurements made by VSWR sensor **32** may be compared to multiple threshold values to identify range R between external object **46** and the corresponding antenna **40**. Curve **110** plots the magnitude of reflection S-parameter S.sub.11 (i.e., |S.sub.11|) as a function of frequency in the absence of external object **46** within threshold range R.sub.TH. As shown by curve **110**, in the absence of external object **46**, |S.sub.11| may have a relatively high value across a frequency band of interest B.

(42) Curve **112** plots |S.sub.11| as a function of frequency when external object **46** is within threshold range R.sub.TH from antenna **40**. As shown by curve **112**, IS iii may have a relatively

low value across frequency band B due to the presence of external object **46**. In general, once external object **46** is within threshold range R.sub.TH, |S.sub.11| will continue to decrease, as shown by arrow **114**, as the object approaches the corresponding antenna **40**. Control circuitry **14** may gather VSWR values using VSWR sensor **32** (e.g., IS ill values such as those shown by curves **110** and **112**) and may process the gathered VSWR values to identify range R when external object **46** is within threshold range R.sub.TH (e.g., by comparing the gathered |S.sub.11| values to one or more threshold levels TH).

(43) For example, when the measured |S.sub.11| value is less than a first threshold TH**0**, control circuitry **14** may determine (e.g., identify, deduce, estimate, etc.) that external object **46** is located at a first range R from antenna **40** (e.g., within threshold range R.sub.TH), when the measured |S.sub.11| is value less than a second threshold TH**1**, control circuitry **14** may determine that external object **46** is located at a second range R from antenna **40** that is closer than the first range, when the measured |S.sub.11| value is less than a third threshold TH**2**, control circuitry **14** may determine that external object **46** is located at a third range R from antenna **40** that is closer than the second range, etc. Beyond threshold range R.sub.TH, |S.sub.11| will exhibit no change or a negligible change in response to changes in distance between antenna **40** and external object **46**. At these relatively far distances, the long range spatial ranging circuitry in communications circuitry **26** (FIG. **1**) may be used to detect the presence, position (e.g., range R), and/or velocity of external object **46**.

(44) The example of FIG. **3** in which control circuitry **14** identifies the range R between a given antenna **40** and external object **46** based on the magnitude of the VSWR measurements (e.g., |S.sub.11| measurements) performed using that antenna **40** is merely illustrative. Additionally or alternatively, control circuitry **14** may identify range R based on the phase of the S.sub.11 measurements. Additionally or alternatively, control circuitry **14** may identify range R based on variations in the VSWR measurements over time.

(45) FIG. **4** is a plot of different reflection coefficient (return loss) magnitude measurements (|S.sub.11| values) that may be made by a given VSWR sensor **32** as a function of time in the presence an external object **46** at different distances (ranges R) from the corresponding antenna **40**. Points **116** of FIG. **4** illustrate |S.sub.11| measurements that may be made by VSWR sensor **32** at sampling times T**0**-T**3** in the presence of an external object (e.g., an animate object) at a first range R from antenna **40** (e.g., within threshold range R.sub.TH). The external object may be, for example, a body part such as a hand, finger, or head. As shown by points **116**, there is a relatively high amount of variation in |S.sub.11| as a function of time in the presence of the external object at the first range R from antenna **40** (e.g., due to minute movements of the external object relative to static/inanimate objects such as a removable device case). Points **118** illustrate |S.sub.11| measurements that may be made by VSWR sensor **32** at times T**0-**T**3** in the presence of the external object at a second range R from antenna **40** that is closer than the first range. As shown by points **118**, there is even more variation in |S.sub.11| as a function of time in the presence of the external object at the second range R from antenna **40** (e.g., because minute movements of the external object produce a greater variation in the impedance loading of antenna **40** and thus the gathered VSWR measurements at closer ranges).

(46) Control circuitry **14** may identify (e.g., detect, produce, compute, calculate, estimate, etc.) variations in the |S.sub.11| measurements over time to identify the range between antenna **40** and the external object (e.g., by comparing the identified variation to one or more threshold variation levels). Control circuitry **14** may perform range detection in this way based on any desired metric for the variation of VSWR (e.g., |S.sub.11|) measurements over time. For example, control circuitry **14** may perform range detection based on the difference between the maximum |S.sub.11| value and the minimum |S.sub.11| value measured at each of the sampling times. For points **116**, control circuitry **14** may identify (e.g., compute, calculate, generate, determine, etc.) a first difference value Δ**1** that is equal to the difference between the maximum |S.sub.11| value B of points **116** (e.g., as

measured at time T**1**) and the minimum |S.sub.11| value C of points **116** (e.g., as measured at time T**2**). Similarly, for points **118**, control circuitry **14** may identify a second difference value Δ**2** that is equal to the difference between the maximum |S.sub.11| value A of points **116** (e.g., as measured at time T**1**) and the minimum |S.sub.11| value D of points **118** (e.g., as measured at time T**0**). Difference value Δ**2** is greater than distance value Δ**1** and is therefore indicative of external object **46** being located at a closer range to antenna **40** than when distance value Δ**1** is measured.
(47) The example of FIG. **4** is merely illustrative. Points **116** and **118** may have other values in practice. In the example of FIG. **4**, four sampling times T**0**-T**3** are used to identify variations in |S.sub.11| for performing animate object detection. This is merely illustrative and, in general, any desired number of sampling times may be used to identify variations in |S.sub.11| for performing animate object detection. Each sampling time may be separated by 10 ms, 20 ms, 1-20 ms, more than 20 ms, 10-50 ms, or any other desired period. The sampling times need not be evenly spaced.
(48) Curve **120** of FIG. **5** shows one example of how variation in |S.sub.11| may be correlated with the range between external object **46** and antenna **40**. If desired, control circuitry **14** may compare the identified variation in the VSWR measurements (e.g., difference value Δ) to curve **120** to identify the corresponding range R between external object **46** and antenna **40**. As shown by curve **120**, control circuitry **14** may determine that external object **46** is located at first range R**1** when difference value Δ**1** is measured (e.g., by identifying the horizontal coordinate on curve **120** corresponding to difference value Δ**1**) and may determine that that external object **46** is located at second range R**2** when difference value Δ**2** is measured. This is merely illustrative and, if desired, control circuitry **14** may identify range R by comparing the measured difference value Δ to one or more threshold difference values, by comparing difference value Δ to entries in a lookup table, database, or other data structure, etc. Curve **120** may be stored on device **10** (e.g., during factory calibration, manufacture, assembly, testing, etc.). The example of FIG. **5** is merely illustrative and, in practice, curve **120** may have other shapes.
(49) The example of FIGS. **4** and **5** is merely illustrative and, in general, control circuitry **14** may identify any desired metric of variance in |S.sub.11| for comparison to one or more threshold values in identifying the range R between external object **46** and antenna **40**. As other examples, control circuitry **14** may identify the mean and variance of the |S.sub.11| measurements over time, the rate of change of |S.sub.11| measurements over time, and/or any other desired variation metrics for comparison to one or more threshold values for identifying range R.
(50) In summary, control circuitry **14** may use VSWR measurements (e.g., |S.sub.11| values) measured using VSWR sensor **32** or variations in the VSWR measurements (e.g., variations in the |S.sub.11| values over time) gathered/performed using VSWR sensor **32** to detect (e.g., identify, determine, estimate, compute, calculate, deduce, etc.) the range R between external object **46** and the corresponding antenna **40**. Control circuitry **14** may process the range R between external object **46** and each antenna **40** in the set of N antennas **40** having a corresponding VSWR sensor **32** to identify the angular location of external object **46**.
(51) FIG. **6** illustrates one example of how the angular location of external object **46** may be defined. External object **46** is illustrated as a human finger herein as an example. This is merely illustrative and, in general, external object **46** may be other body parts of the user of device **10**, other humans or animals, furniture, walls, ceilings, the ground, a peripheral device or accessory such as a gaming controller, user interface/input device, or headset, and/or any other object external to device **10**.
(52) In the example of FIG. **6**, the control circuitry on device **10** (e.g., control circuitry **14** of FIG. **1**) uses a spherical coordinate system to determine the location and orientation of external object **46** relative to a (lateral) surface **122** of device **10**. Surface **122** may, for example, be a surface of a housing wall for device **10** (e.g., housing **12** of FIG. **1**), a surface of a cover layer for device **10** (e.g., a dielectric cover layer), a surface of a display on or mounted to the housing, the surface of an antenna module on or within device **10**, or any other desired surface on, at, or within device **10**.

Surface **122** need not be planar.

(53) In this type of coordinate system, control circuitry **14** may process the range R between two or more antennas **40** (e.g., as identified using VSWR measurements gathered using the two or more antennas as described above in connection with FIGS. **2-5**) to determine (e.g., calculate, compute, detect, estimate, deduce, generate, etc.) an azimuth angle θ and/or an elevation angle φ that characterize (identify) the angular location of external object **46** relative to a point P on surface **122** (or some other reference surface). When viewed in the −Z direction, point P may be laterally located between two or more of the antennas **40** in device **10** having a corresponding VSWR sensor **32** (for example).

(54) In identifying the angular location of external object **46** (e.g., a spherical coordinate value (θ, φ), sometimes referred to herein as angle-of-arrival), control circuitry **14** may define a reference plane at lateral surface **122** and a reference vector such as reference vector **126**. Reference vector **126** may lie within the reference plane (e.g., lateral surface **122**).

(55) As shown in FIG. **6**, external object **46** may be separated from point P by range R (e.g., where range R is the magnitude of a positional vector extending from point P to external object **46**). The elevation angle φ (sometimes referred to as altitude) of external object **46** may be measured as the angle between the positional vector extending from point P to external object **46** and the reference plane (e.g., lateral surface **122**). The azimuth angle θ of external object **46** may be measured as the angle of external object **46** around the reference plane (e.g., the angle between reference vector **126** and vector **128**, which is the horizontal projection of the positional vector extending from point P to external object **46** within the reference plane). In the example of FIG. **6**, the azimuth angle θ and elevation angle φ of external object **46** are each greater than 0°.

(56) If desired, other axes may be used to define reference vector **126** (e.g., reference vector **126** may point in any direction). Other angles may be used to characterize the angular location of external object **46** (e.g., the angle between the normal vector (axis) **124** of surface **122** and the positional vector extending from point P to external object **46**, which is equal to 90°−φ, other angles, etc.). The example of FIG. **6** in which the angular position of external object **46** is characterized using spherical coordinates (e.g., as an angular position (θ, φ)) is merely illustrative. In general, control circuitry **14** may characterize or identify the angular location of external object **46** using any desired coordinate system (e.g., rectangular/Cartesian coordinates, polar coordinates, cylindrical coordinates, other coordinate systems, etc.) about any desired reference axes. The reference plane may be arbitrarily selected if desired and need not coincide with the presence of a surface of device **10** such as surface **122**.

(57) FIG. **7** is a flow chart of illustrative operations that may be performed by device **10** to identify the angular location of external object **46** using VSWR values gathered using multiple antennas **40** on device **10** (e.g., a set of N antennas **40** each having a respective VSWR sensor **32** communicatively coupled thereto along a respective transmit path **34**). The number N may be two, three, four, five, six, seven, eight, or more than eight, as examples. Each of the N antennas **40** may be disposed at different points on/across device **10**.

(58) At operation **130**, control circuitry **14** may control wireless circuitry **24** to transmit test signals sigtx (FIG. **2**) over each of the N antennas **40**. Test signals sigtx may be transmitted over each of the N antennas **40** concurrently (e.g., simultaneously) or sequentially (in series). Test signals sigtx may be transmitted at a single carrier frequency or over a range/sweep of frequencies. The N antennas **40** may begin transmitting test signals sigtx once one or more of the VSWR sensors **32** has already detected that external object **46** has passed within threshold range R.sub.TH of the corresponding antenna **40** or in response to any desired trigger condition (e.g., an application or software call on control circuitry **14**, once VSWR measurements such as |S.sub.11| measurements reach a predetermined threshold value, etc.).

(59) As another example, the control circuitry **14** may perform operation **130** once device **10** has determined that gathered wireless performance metric data has fallen outside of a predetermined

range. In this example, wireless circuitry **24** may gather wireless performance metric data associated with the radio-frequency performance of antenna(s) **40**. The wireless performance metric data may include signal-to-noise ratio (SNR) data, receive signal strength indicator (RSSI) data, or any other desired performance metric data gathered during the transmission of radio-frequency signals **38**, the transmission of radio-frequency signals **42**, the reception of radio-frequency signals **38**, and/or the reception of reflected signals **44** of FIG. **1**, for example. Control circuitry **14** may compare the gathered wireless performance metric data with a predetermined range of wireless performance metric values associated with satisfactory radio-frequency performance and/or the operation of wireless circuitry **24** in the absence of external objects within threshold range R.sub.TH (e.g., a predetermined range of satisfactory RSSI values, SNR values, etc.). The predetermined range of wireless performance metric values may be characterized by an upper threshold limit or value and/or a lower threshold limit or value.

(60) In this example, the wireless performance metric data may serve as a coarse indicator for whether external object **46** is within threshold range R.sub.TH. For example, if external object **46** is within range R.sub.TH, external object **46** may partially block or cover one or more antennas **40** (thereby preventing the antenna from properly receiving radio-frequency signals), may undesirably load or detune one or more antennas **40** in device **10**, etc. When the gathered wireless performance metric data falls outside of the predetermined range, this may be indicative of the potential presence of external object **46** within threshold range R.sub.TH. However, when the gathered wireless performance metric data falls within the predetermined range, this may indicate that it is very unlikely that there is an external object present within threshold range R.sub.TH (e.g., because wireless circuitry **24** is performing nominally as expected in the absence of an external object within threshold range R.sub.TH). If the gathered wireless performance metric data falls within the predetermined range (thereby indicating that there is no external object within threshold range R.sub.TH), VSWR sensor(s) **32** may gather background VSWR measurements for performing background cancellation if desired. In general, operation **130** may be performed in response to any desired trigger condition.

(61) At operation **132**, control circuitry **14** may use the respective VSWR sensor **32** (e.g., measurement circuitry **70** of FIG. **2**) coupled to each of the N antennas **40** to perform at least N VSWR measurements (e.g., S.sub.11 values, |S.sub.11| values, time-variations in the |S.sub.11| values such as difference values Δ of FIGS. **4** and **5**, etc.) for each of the N antennas **40** using the transmitted test signals sigtx. Control circuitry **14** may perform the VSWR measurements for each of the N antennas **40** concurrently (e.g., when the N antennas **40** transmit test signals sigtx concurrently) or sequentially (e.g., when the N antennas **40** transmit test signals sequentially). Operation **132** may, for example, be performed concurrently with operation **130**. In examples where the VSWR measurements are performed sequentially, two or more (e.g., all) of the N antennas **40** may share a single transmit path **34** and VSWR sensor **32** if desired.

(62) At operation **134**, control circuitry **14** may process the VSWR measurements for each of the N antennas **40** to identify (e.g., determine, detect, estimate, calculate, compute, deduce, etc.) the respective range R between each of the N antennas **40** and external object **46**. Control circuitry **14** may identify ranges R by comparing the VSWR measurements to one or more threshold values. For example, control circuitry **14** may identify ranges R by comparing |S.sub.11| values gathered using each of the N antennas **40** to threshold values TH of FIG. **3**, by comparing difference values Δ of FIG. **4** to one or more threshold values or to curve **120** of FIG. **5**, by comparing the phases of the VSWR measurements to one or more threshold values, etc. Control circuitry **14** need not use the same method to calculate VSWR for each of the N antennas **40** (e.g., control circuitry **14** may identify range R using variations in the VSWR measurements over time for some of the N antennas **40** while identifying range R using |S.sub.11| values for others of the N antennas **40**, etc.).

(63) If desired, control circuitry **14** may identify range R while also performing VSWR background cancellation. For example, control circuitry **14** may use the VSWR sensor(s) **32** to gather

background VSWR measurements in the absence of other external objects within threshold range R.sub.TH from the corresponding antenna(s) **40** (e.g., where the background VSWR measurements also take into account the presence of the removable device case). Control circuitry **14** may then use the background VSWR measurements to perform background cancellation on subsequent VSWR measurements (e.g., as performed at operation **132**) that are gathered in the presence of external object **46** within threshold range R.sub.TH (e.g., by subtracting the background VSWR measurements from the subsequent VSWR measurements).

(64) At operation **136**, control circuitry **14** may process each of the N identified ranges R (e.g., the range R identified between each of the N antennas **40** and external object **46**) to identify (e.g., determine, calculate, estimate, deduce, generate, triangulate, resolve, etc.) the angular location of external object **46** relative to surface **122** of device **10** (FIG. **6**) or any other desired reference plane. Control circuitry **14** may identify the angular location of external object **46** (e.g., a point (θ, φ) in spherical coordinates or any other desired coordinate system) by performing geometric calculations based on (using) the identified ranges R between the N antennas **40** and external object **46** and the known (predetermined) separation/spacing between each of the N antennas. Each range R may, for example, be the radius of a sphere of potential locations for device **10** centered on the corresponding antenna **40**. Control circuitry **14** may identify the location of external object **46** as the location/point where each of the N spheres intersect in space. Control circuitry **14** may then identify the angular location of external object **46** as the angle of a vector extending from any desired point on lateral surface **122** (e.g., point P of FIG. **6**) to the location/point where each of the N spheres intersect in space (e.g., angles relative to any desired vectors such as vectors **126**, **128**, and/or **124** of FIG. **6** or other vectors and in any desired coordinate system). In spherical coordinates, this angle may be characterized by a spherical point (θ, φ), for example. Additionally or alternatively, control circuitry **14** may identify the angular location using a lookup table, database, or other data structure that maps different range values R for each of the N antennas to different angular locations for external object **46** (in any desired angular coordinate system about any desired reference points or reference vectors). The lookup table, database, or other data structure may be populated during design, manufacture, assembly, testing, or calibration of device **10** and/or may be populated/updated during operation of device **10** by an end user.

(65) At operation **138**, control circuitry **14** may perform any desired processing operations based on the identified angular location of external object **46**. For example, at operation **140**, control circuitry **14** may adjust the transmit power level or maximum transmit power level of one or more of the antennas **40** on device **10** based on the angular location of external object **46** (e.g., control circuitry **14** may increase the transmit power level or the maximum transmit power level of antennas **40** that are relatively far from external object **46** and/or may decrease the transmit power level or maximum transmit power level of antennas **40** that are relatively close to external object **46**). If desired, control circuitry **14** may disable or activate antennas **40** based on the identified angular location (e.g., control circuitry **14** may switch antennas **40** that are too close to external object **46** out of use). These techniques may, for example, help to ensure that device **10** continues to satisfy regulatory limits on radio-frequency energy exposure (e.g., SAR/MPE limits).

(66) As another example, at operation **142**, control circuitry **14** may adjust the angle of a signal beam produced by a phased antenna array of the antennas **40** in device **10** (e.g., the N antennas **40** used to gather VSWR measurements and/or other antennas **40**) based on the identified angular location of external object **46**. For example, control circuitry **14** may adjust (steer) the signal beam around the identified angular location (e.g., to point the signal beam in a different angle than the identified angular location). This may prevent the signal beam from overlapping the external object, thereby helping device **10** to satisfy regulatory limits on radio-frequency energy exposure while also allowing device **10** to continue to perform wireless operations over the signal beam without the external object blocking the signal beam.

(67) As yet another example, at operation **144**, control circuitry **14** may identify a user input action

such as a gesture action based on the identified angular location. Control circuitry **14** may identify a particular user input or gesture corresponding to the identified angular location and/or corresponding to particular changes in angular location of external object **46** over time (e.g., over multiple iterations of operations **130-136** of FIG. **7**). The user input or gesture may, for example, form a user input used by software applications running on device **10** to perform any desired processing tasks, operations, or functions. The gestures may, for example, be used to control, perform, or coordinate on-screen actions displayed on a display for device **10** using the software applications.

(68) The example of FIG. **7** is merely illustrative. Operations **140**, **142**, and/or **144** may be omitted. Control circuitry **14** may perform any other desired processing operations or device functions based on the identified angular location of external object **46** and/or the identified angular location of external object **46** over time (e.g., over multiple iterations of the operations of FIG. **7**).

(69) FIG. **8** is a top view showing one example of how N=4 antennas **40** may be used to identify the angular location of external object **46**. In the example of FIG. **8**, antennas **40-1**, **40-2**, **40-3**, and **40-4** may each be coupled to a respective transmit path **34** having a respective VSWR sensor **32**. In examples where the N antennas gather VSWR measurements sequentially, two or more of the antennas may share a single transmit path **34** and VSWR sensor **32**. Antennas **40-1**, **40-2**, **40-3**, and/or **40-4** may also be used to convey wireless communications data and/or to perform long range spatial ranging for communications circuitry **26** of FIG. **1**. If desired, some or all of antennas **40-1**, **40-2**, **40-3**, and **40-4** may form part of a phased antenna array (e.g., antennas **40-1**, **40-2**, **40-3**, and **40-4** may be a four-element phased antenna array).

(70) As shown in FIG. **8**, antennas **40-1**, **40-2**, **40-3**, and **40-4** may be formed on or within a substrate such as substrate **146**. Substrate **146** may be a printed circuit such as a rigid printed circuit board or flexible printed circuit, may be a plastic, ceramic, or glass substrate, may be a housing wall or cover layer for device **10**, may be a portion of a display for device **10**, or may be any other desired dielectric material. This example is merely illustrative and, if desired, each antenna may be disposed on a respective substrate **146**, the antennas may be divided between two or more substrates **146**, or one or more of the antennas may be disposed in device **10** without a substrate. The uppermost surface of substrate **146** may, for example, form surface **122** of FIG. **6**.

(71) During angular detection operations, antennas **40-1**, **40-2**, **40-3**, and **40-4** may each transmit test signals sigtx (e.g., while processing operation **130** of FIG. **7**). The VSWR sensor(s) **32** coupled to antennas **40-1**, **40-2**, **40-3**, and **40-4** may perform VSWR measurements using the test signals sigtx transmitted by each of the antennas (e.g., while processing operation **132** of FIG. **7**). Control circuitry **14** may process the VSWR measurement(s) performed using antenna to identify the range R**1** between antenna **40-1** and external object **46**, may process the VSWR measurement(s) performed using antenna **40-2** to identify the range R**2** between antenna and external object **46**, may process the VSWR measurement(s) performed using antenna to identify the range R**3** between antenna **40-3** and external object **46**, and may process the VSWR measurement(s) performed using antenna **40-4** to identify the range R**4** between antenna and external object **46**.

(72) The top view of FIG. **8** shows the lateral projections of ranges R**1**-R**4** in the X-Y plane. FIG. **9** is a side view of the antennas **40-1**, **40-2**, **40-3**, and **40-4** on substrate **146** (e.g., as taken in the direction of arrow **148** of FIG. **8**). FIG. **9** shows the projections of ranges R**1**-R**4** in the X-Z plane (e.g., ranges R**1**-R**4** may be the magnitudes of three-dimensional position vectors extending from antennas **40-1**, **40-2**, **40-3**, and **40-4** to external object **46**, respectively). Range R**1** may correspond to the radius of a sphere of potential locations for external object **46** that is centered on antenna **40-1**. Range R**2** may correspond to the radius of a sphere of potential locations for external object **46** that is centered on antenna **40-2**. Range R**3** may correspond to the radius of a sphere of potential locations for external object **46** that is centered on antenna **40-3**. Range R**4** may correspond to the radius of a sphere of potential locations for external object **46** that is centered on antenna **40-4**.

(73) Control circuitry **14** may process ranges R**1**-R**4** to identify the angular location of external

object **46** while processing operation **136** of FIG. **7** (e.g., by identifying the angular position of the point/location where each of the spheres corresponding to ranges R**1**-R**4** intersect, by comparing ranges R**1**-R**4** to a lookup table of angular locations, etc.). For example, control circuitry **14** may identify the angle θ at point P to external object **46** relative to reference vector **126** (FIG. **8**) and the angle φ at point P to external object **46** relative to the lateral surface of substrate **146** (e.g., lateral surface **122** of FIG. **6**) in spherical coordinates. Point P may be located between (e.g., equidistant from) antennas **40-1**, **40-2**, **40-3**, and **40-4** or may be at any other desired location on the lateral surface of substrate **146**. This is merely illustrative and, in general, control circuitry **14** may identify the angular location of external object **46** using any desired coordinate system and with respect to any desired location (e.g., point P) on device **10**. The example of FIGS. **8** and **9** in which N=4 antennas **40** are used to identify the angular location of external object **46** is merely illustrative and, in general, N may have other values greater than two. The N antennas may be arranged in any desired pattern (e.g., in a two-dimensional array pattern, a one-dimensional array pattern, a pattern of concentric rings, etc.) and may be formed using any desired type of antenna resonating elements.

(74) FIG. **10** is a side view showing how the angular location of external object **46** may be used to perform beam steering operations (e.g., while processing operation **142** of FIG. **7**). As shown in FIG. **10**, device **10** may include a phased antenna array **156** (sometimes referred to herein as array **156**, antenna array **156**, or array **156** of antennas **40**). Phased antenna array **156** may include M antennas **40** such as a first antenna **40-1**, an Mth antenna **40**-M, etc. The antennas in phased antenna array **156** may be disposed on substrate **146**, on another substrate, or may be distributed across two or more substrates. Phased antenna array **156** may be coupled to radio-frequency transmission line paths **150** (e.g., radio-frequency transmission line paths used to form transmit path(s) **34** and/or receive path(s) **36** of FIG. **1**). For example, a first antenna **40-1** in phased antenna array **156** may be coupled to a first radio-frequency transmission line path **150-1**, an Mth antenna **40**-M in phased antenna array **156** may be coupled to an Mth radio-frequency transmission line path **150**-M, etc. Some, none, or all of the M antennas in phased antenna array **40** may be among the N antennas **40** used to identify the angular location of external object **46**. While antennas **40** are described herein as forming a phased antenna array, the antennas **40** in phased antenna array **156** may sometimes also be referred to as collectively forming a single phased array antenna (e.g., where each antenna **40** in the phased array antenna forms an antenna element or radiator of the phased array antenna).

(75) Radio-frequency transmission line paths **150** may each be coupled to transceiver circuitry such as a 5G NR transceiver in communications circuitry **26** of FIG. **1**. Each radio-frequency transmission line path **150** may include one or more radio-frequency transmission lines, a positive signal conductor, and a ground signal conductor. The positive signal conductor may be coupled to a positive antenna feed terminal on an antenna resonating element of the corresponding antenna **40**. The ground signal conductor may be coupled to a ground antenna feed terminal on an antenna ground for the corresponding antenna **40**.

(76) The antennas **40** in phased antenna array **156** may be arranged in any desired number of rows and columns or in any other desired pattern (e.g., the antennas need not be arranged in a grid pattern having rows and columns). During signal transmission operations, radio-frequency transmission line paths **150** may be used to supply signals (e.g., radio-frequency signals such as millimeter wave and/or centimeter wave signals) from the transceiver in communications circuitry **26** (FIG. **1**) to phased antenna array **156** for wireless transmission. During signal reception operations, radio-frequency transmission line paths **150** may be used to convey signals received at phased antenna array **156** (e.g., from external wireless equipment **28** of FIG. **1**) to the transceiver in communications circuitry **26**.

(77) The use of multiple antennas **40** in phased antenna array **156** allows radio-frequency beam forming arrangements (sometimes referred to herein as radio-frequency beam steering arrangements) to be implemented by controlling the relative phases and magnitudes (amplitudes) of

the radio-frequency signals conveyed by the antennas. In the example of FIG. **10**, the antennas **40** in phased antenna array **156** each have a corresponding radio-frequency phase and magnitude controller **152** (e.g., a first phase and magnitude controller **152-1** interposed on radio-frequency transmission line path **150-1** may control phase and magnitude for radio-frequency signals handled by antenna **40-1**, an Mth phase and magnitude controller **152**-M interposed on radio-frequency transmission line path **150**-M may control phase and magnitude for radio-frequency signals handled by antenna **40**-M, etc.).

(78) Phase and magnitude controllers **152** may each include circuitry for adjusting the phase of the radio-frequency signals on radio-frequency transmission line paths **150** (e.g., phase shifter circuits) and/or circuitry for adjusting the magnitude of the radio-frequency signals on radio-frequency transmission line paths **150** (e.g., power amplifier and/or low noise amplifier circuits). Phase and magnitude controllers **152** may sometimes be referred to collectively herein as beam steering or beam forming circuitry (e.g., beam steering circuitry that steers the beam of radio-frequency signals transmitted and/or received by phased antenna array **156**).

(79) Phase and magnitude controllers **152** may adjust the relative phases and/or magnitudes of the transmitted signals that are provided to each of the antennas in phased antenna array **156** and may adjust the relative phases and/or magnitudes of the received signals that are received by phased antenna array **156**. Phase and magnitude controllers **152** may, if desired, include phase detection circuitry for detecting the phases of the received signals that are received by phased antenna array **156**. The term “beam,” “signal beam,” “radio-frequency beam,” or “radio-frequency signal beam” may be used herein to collectively refer to wireless signals that are transmitted and received by phased antenna array **156** in a particular direction. The signal beam may exhibit a peak gain that is oriented in a particular beam pointing direction at a corresponding beam pointing angle (e.g., based on constructive and destructive interference from the combination of signals from each antenna in the phased antenna array). The term “transmit beam” may sometimes be used herein to refer to radio-frequency signals that are transmitted in a particular direction whereas the term “receive beam” may sometimes be used herein to refer to radio-frequency signals that are received from a particular direction.

(80) If, for example, phase and magnitude controllers **152** are adjusted to produce a first set of phases and/or magnitudes for transmitted radio-frequency signals, the transmitted signals will form a transmit beam that is oriented in a first direction such as the direction of external object **46**. If, however, phase and magnitude controllers **152** are adjusted to produce a second set of phases and/or magnitudes for the transmitted signals, the transmitted signals will form a transmit beam as shown by beam **160** that is oriented in direction **158**, which points away from external object **46**. Similarly, if phase and magnitude controllers **152** are adjusted to produce the first set of phases and/or magnitudes, radio-frequency signals (e.g., radio-frequency signals in a receive beam) may be received from the direction external object **46**. If phase and magnitude controllers **152** are adjusted to produce the second set of phases and/or magnitudes, radio-frequency signals may be received from direction **158**, as shown by beam **160**.

(81) Each phase and magnitude controller **152** may be controlled to produce a desired phase and/or magnitude based on a corresponding control signal **154** received from control circuitry **14** of FIG. **1** (e.g., the phase and/or magnitude provided by phase and magnitude controller **152-1** may be controlled using control signal **154-1**, the phase and/or magnitude provided by phase and magnitude controller **152**-M may be controlled using control signal **154**-M, etc.). If desired, control circuitry **14** may actively adjust control signals **154** in real time to steer the transmit or receive beam in different desired directions (e.g., to different desired beam pointing angles) over time. In the example of FIG. **10**, beam steering is shown as being performed over a single degree of freedom for the sake of simplicity (e.g., towards the left and right on the page of FIG. **10**). However, in practice, the beam may be steered over two or more degrees of freedom (e.g., in three dimensions, into and out of the page and to the left and right on the page of FIG. **10**) or in a single

degree of freedom (e.g., when the antennas **40** in phased antenna array are arranged in a one-dimensional pattern). Phased antenna array **156** may have a corresponding field of view over which beam steering can be performed (e.g., in a hemisphere or a segment of a hemisphere over the phased antenna array). If desired, device **10** may include multiple phased antenna arrays that each face a different direction to provide coverage from multiple sides of the device.

(82) While processing operation **136** of FIG. **7**, control circuitry **14** may determine that external object **46** is at direction (angular location) J with respect to phased antenna array **156**. While processing operation **142** of FIG. **7**, control circuitry **14** may adjust phase and magnitude controllers **152** to steer signal beam **160** in direction **158** (e.g., away from direction J) so the signal beam does not overlap external object **46**. This may help to ensure that phased antenna array **156** continues to comply with regulations on RF exposure and/or to ensure that phased antenna array **156** is able to convey wireless communications data and/or perform spatial ranging operations despite the presence of external object **46** in proximity to phased antenna array **156**.

(83) If desired, the N antennas **40** used to identify the angular location of external object **46** may be distributed across two or more substrates. FIG. **11** is a top view of device **10** showing one example of how the N antennas **40** may be distributed across two substrates.

(84) As shown in FIG. **11**, the N antennas **40** used to identify the angular location of external object **46** may include a first set of antennas **40** on a first substrate **146**A and a second set of antennas **40** on a second substrate **146**B disposed within housing **12** of device **10**. The first set of antennas may, for example, be arranged in a one-dimensional pattern on substrate **146**A whereas the second set of antennas are arranged in a one-dimensional pattern on substrate **146**B. Because a single one-dimensional array of antennas may be insufficient to fully resolve ambiguities in the angular location of external object **46**, the second set of antennas **40** on substrate **146**B may be oriented perpendicular to the first set of antennas **40** on substrate **146**A (e.g., the antennas **40** on substrate **146**A may be disposed along a first axis, the antennas **146**B may be disposed along a second axis, and the second axis may be oriented perpendicular to the first axis). The first and the second sets of antennas may then be able to resolve the correct angular location of external object **46**. The first set of antennas **40** and the second set of antennas **40** may radiate through a rear face of device **10** (e.g., a face of device **10** opposite to a display for device **10**) or may radiate through the front face of device **10**. Substrates **146**A and **146**B may, for example, be sufficiently narrow so as to allow the N antennas distributed across substrate **146**A and **146**B to perform VSWR measurements through an inactive area of a display on the front face of device (e.g., an area of the display that is overlapped by a dielectric cover layer and that is laterally interposed between an active light-emitting area of the display and peripheral conductive housing structures for device **10**). The first and second sets of antennas may form respective one-dimensional phased antenna arrays **156** if desired. Additionally or alternatively, the N antennas may be disposed in a two-dimensional array pattern on one or more substrates **146**.

(85) These examples are merely illustrative. Some or all of the N antennas **40** need not be disposed on substrate **146** or arranged in any array pattern. More generally, the N antennas **40** used to measure the angular location of external object **46** may be distributed across any desired locations on device **10**. FIG. **12** is a top view showing illustrative locations for distributing some or all of the N antennas **40** used to measure the angular location of external object **46**.

(86) As shown in FIG. **12**, one or more of the N antennas **40** may be located within one or more regions **164** on or within device **10** such as region **164-1** at the top-left corner of device **10**, region **164-2** at the top-right corner of device **10**, region **164-3** at the bottom-left corner of device region **164-4** at the bottom-right corner of device **10**, one or more regions **164-5** within a central region of device **10**, and/or one or more regions **164-6** laterally interposed between an active area of a display for device **10** and housing **12**.

(87) Separating two or more of the N antennas **40** by relatively large distances and increasing the number N of antennas **40** used to perform VSWR measurements may increase the resolution with

which control circuitry **14** is able to determine the angular location of external object **46**. Control circuitry **14** may determine the angular location of external object **46** with an angular resolution of as fine as 1-2°, for example. In the example of FIG. **12**, one or more of the N antennas **40** located in regions **164-1**, **164-2**, **164-3**, and **164-4** may have radiating elements (e.g., antenna resonating element arms) formed from conductive segments of housing **12** (e.g., peripheral conductive housing structures that run around the lateral periphery of device **10**) that are separated/defined by dielectric-filled gaps **162** in housing **12**. The antennas formed from conductive portions of housing **12** may also be used to convey cellular telephone data, WLAN data, GPS data, etc. The example of FIG. **12** is merely illustrative. In general, housing **12** may have any desired shape.
(88) The methods and operations described above in connection with FIGS. **1-12** may be performed by the components of device **10** using software, firmware, and/or hardware (e.g., dedicated circuitry or hardware). Software code for performing these operations may be stored on non-transitory computer readable storage media (e.g., tangible computer readable storage media) stored on one or more of the components of device **10** (e.g., storage circuitry **16** of FIG. **1**). The software code may sometimes be referred to as software, data, instructions, program instructions, or code. The non-transitory computer readable storage media may include drives, non-volatile memory such as non-volatile random-access memory (NVRAM), removable flash drives or other removable media, other types of random-access memory, etc. Software stored on the non-transitory computer readable storage media may be executed by processing circuitry on one or more of the components of device **10** (e.g., processing circuitry **18** of FIG. **1**, etc.). The processing circuitry may include microprocessors, central processing units (CPUs), application-specific integrated circuits with processing circuitry, or other processing circuitry. The components of FIGS. **1** and **2** may be implemented using hardware (e.g., circuit components, digital logic gates, etc.) and/or using software where applicable.
(89) Device **10** may gather and/or use personally identifiable information. It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.
(90) The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. An electronic device comprising: a first antenna configured to transmit a first radio-frequency signal; a first voltage standing wave ratio (VSWR) sensor communicatively coupled to the first antenna and configured to perform a first VSWR measurement using the first radio-frequency signal; a second antenna configured to transmit a second radio-frequency signal; a second VSWR sensor communicatively coupled to the second antenna and configured to perform a second VSWR measurement using the second radio-frequency signal; one or more processors configured to detect a gesture based on the first VSWR measurement and the second VSWR measurement; a display, the one or more processors being configured to coordinate, based on the detected gesture, an on-screen action displayed by the display; a first substrate, the first antenna being disposed on the first substrate; and a second substrate different from the first substrate, the second antenna being disposed on the second substrate.
2. The electronic device of claim 1, further comprising: a third antenna configured to transmit a third radio-frequency signal; and a third VSWR sensor communicatively coupled to the third antenna and configured to perform a third VSWR measurement using the third radio-frequency

signal, the one or more processors being configured to detect the gesture based on the third VSWR measurement.

3. The electronic device of claim 1, further comprising: a phased antenna array configured to convey a beam of radio-frequency signals, the phased antenna array including the first antenna and the second antenna.

4. The electronic device of claim 3, the one or more processors being configured to adjust a direction of the beam based on the detected gesture.

5. The electronic device of claim 1, further comprising: a first phased antenna array configured to convey a first beam of radio-frequency signals, the first phased antenna array including the first antenna; and a second phased antenna array configured to convey a second beam of radio-frequency signals, the second phased antenna array including the second antenna.

6. The electronic device of claim 1, the one or more processors being configured to: detect a first range between the electronic device and an external object based on the first VSWR measurement; detect a second range between the electronic device and the external object based on the second VSWR measurement; and detect the gesture based on the first range and the second range.

7. The electronic device of claim 1, the one or more processors being configured to detect the gesture based on a change in the first VSWR measurement over time and a change in the second VSWR measurement over time.

8. A method of operating an electronic device, the method comprising: transmitting, using a first antenna, a first radio-frequency signal; transmitting, using a second antenna, a second radio-frequency signal; generating, using a first voltage standing wave ratio (VSWR) sensor, a first scattering parameter value based on the first radio-frequency signal; generating, using a second VSWR sensor, a second scattering parameter value based on the second radio-frequency signal; and detecting, using one or more processors, a gesture based on the first scattering parameter value and the second scattering parameter value.

9. The method of claim 8, further comprising: transmitting, using a third antenna, a third radio-frequency signal; and generating, using a third VSWR sensor, a third scattering parameter value based on the third radio-frequency signal, wherein detecting the gesture comprises detecting the gesture based on the third scattering parameter value.

10. The method of claim 8, further comprising: transmitting, using a fourth antenna, a fourth radio-frequency signal; and generating, using a fourth VSWR sensor, a fourth scattering parameter value based on the fourth radio-frequency signal, wherein detecting the gesture comprises detecting the gesture based on the fourth scattering parameter value.

11. The method of claim 8, wherein transmitting the second radio-frequency signal comprises transmitting the second radio-frequency signal concurrent with transmission of the first radio-frequency signal by the first antenna.

12. The method of claim 8, wherein the first scattering parameter value comprises a first reflection coefficient associated with the first antenna and the second scattering parameter value comprises a second reflection coefficient associated with the second antenna.

13. The method of claim 8, further comprising: displaying images using a display; and adjusting, using the one or more processors, the images based on the detected gesture.

14. An electronic device comprising: a phased antenna array that includes first, second, and third antennas, wherein the first and second antennas are aligned along a first axis, the second and third antennas are aligned along a second axis orthogonal to the first axis, the first antenna is configured to transmit a first radio-frequency signal, the second antenna is configured to transmit a second radio-frequency signal, the third antenna is configured to transmit a third radio-frequency signal, and the phased antenna array is configured to form a signal beam in a beam pointing direction; a first voltage standing wave ratio (VSWR) sensor communicatively coupled to the first antenna and configured to perform a first VSWR measurement using the first radio-frequency signal; a second VSWR sensor communicatively coupled to the second antenna and configured to perform a second

VSWR measurement using the second radio-frequency signal; and one or more processors configured to adjust, based on the first VSWR measurement, the second VSWR measurement, and the third radio-frequency signal, the beam pointing direction of the signal beam formed by the phased antenna array.
15. The electronic device of claim 14, the one or more processors being further configured to: identify a location of an external object based on the first VSWR measurement and the second VSWR measurement; and adjust the beam pointing direction to point away from the identified location of the external object.
16. The electronic device of claim 14, further comprising: a third VSWR sensor communicatively coupled to the third antenna and configured to perform a third VSWR measurement using the third radio-frequency signal, the one or more processors being configured to adjust the beam pointing direction based on the third VSWR measurement.
17. The electronic device of claim 16, wherein the first antenna, the second antenna, and the third antenna are disposed on a substrate.
18. The electronic device of claim 14, wherein the signal beam is formed via constructive and destructive interference of at least the first, second, and third radio-frequency signals.